

Unit 5, Lesson 6: Pythagorean Theorem

Benchmark

MA.8.GR.1.1 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.

Learning Targets

- ☐ I can apply the Pythagorean Theorem to find the hypotenuse of a triangle.
- ☐ I can apply the Pythagorean Theorem to find the leg of a triangle.

5.6.1 Warm-Up: Which One Doesn't Belong?

1. Which one doesn't belong?
- A. $3^2 + b^2 = 5^2$

B. $b^2 = 5^2 - 3^2$

C. $3^2 + 5^2 = b^2$

D. $3^2 + 4^2 = 5^2$
2. Explain your thinking.

5.6.2 Refresher: Parts of a Right Triangle and Using Square Roots

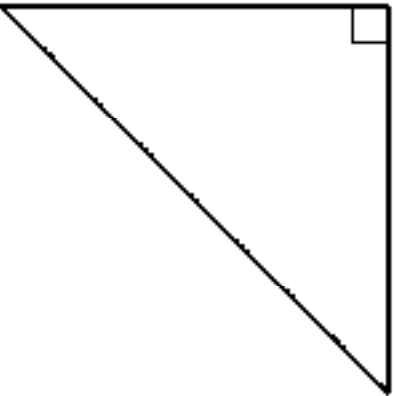


The **hypotenuse** is the longest side of a right triangle; the side opposite the right angle.

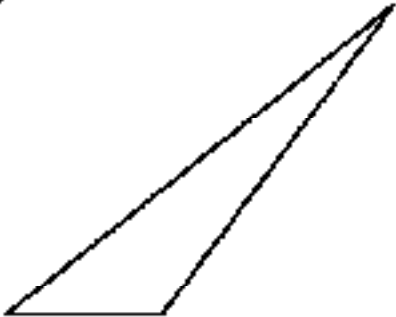
The hypotenuse can be found directly across the right angle.

1. For each right triangle, label the legs a and b and the hypotenuse with c .

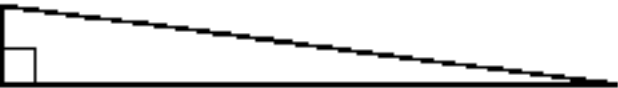
A



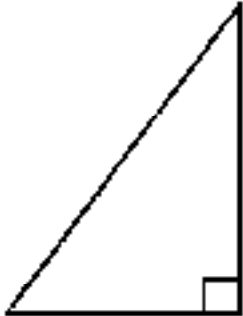
B



C



D



The **Converse of the Pythagorean Theorem** states that if the lengths a , b , and c of the three sides of a triangle satisfy the relationship $a^2 + b^2 = c^2$, then the triangle is a right triangle.

2. Solve for the positive value of the missing variable in each equation. If necessary, approximate your answer.

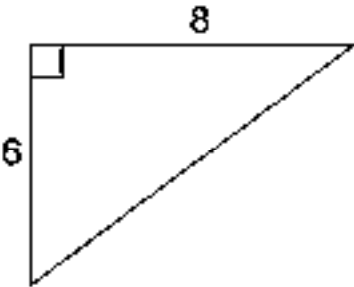
a. $c^2 = 64$

b. $d^2 = 100 + 21$

3. What strategy did you use to find each missing value?

5.6.3 Guided Instruction: Solving for the Hypotenuse Using the Pythagorean Theorem

1. A right triangle with side lengths 6 and 8 is shown.



a. Which side length of the triangle is unknown?

- ☐ hypotenuse
- ☐ legs

b. Which theorem applies specifically to the side lengths of right triangles?

- ☐ Pythagorean Theorem
- ☐ Triangle Inequality Theorem
- ☐ legs

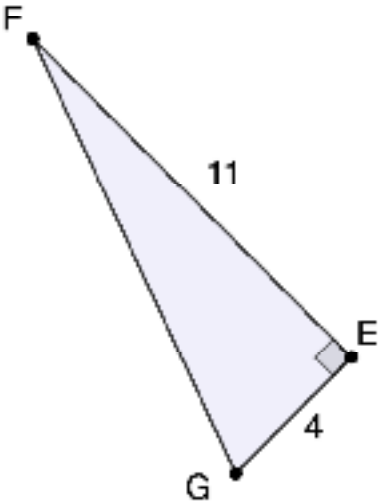
c. Use the Pythagorean Theorem to substitute the given side lengths for the appropriate variables.

$$a^2 + b^2 = c^2$$
$$\underline{\hspace{1cm}}^2 + \underline{\hspace{1cm}}^2 = \underline{\hspace{1cm}}^2$$

d. Solve the equation for the value of the missing variable.

5.6.4 Try It: Finding the Hypotenuse

1. Right triangle FEG is shown with side lengths 4 and 11.



Find the length of the hypotenuse using the Pythagorean Theorem. If the measure of the side length is an irrational value, write an exact solution and an approximate solution.

a. Which side length is unknown?

b. State Pythagorean Theorem.

c. Which variable in the equation is unknown?

d. Substitute values into equation.

e. Algebraically solve for missing variable.

f. Solutions

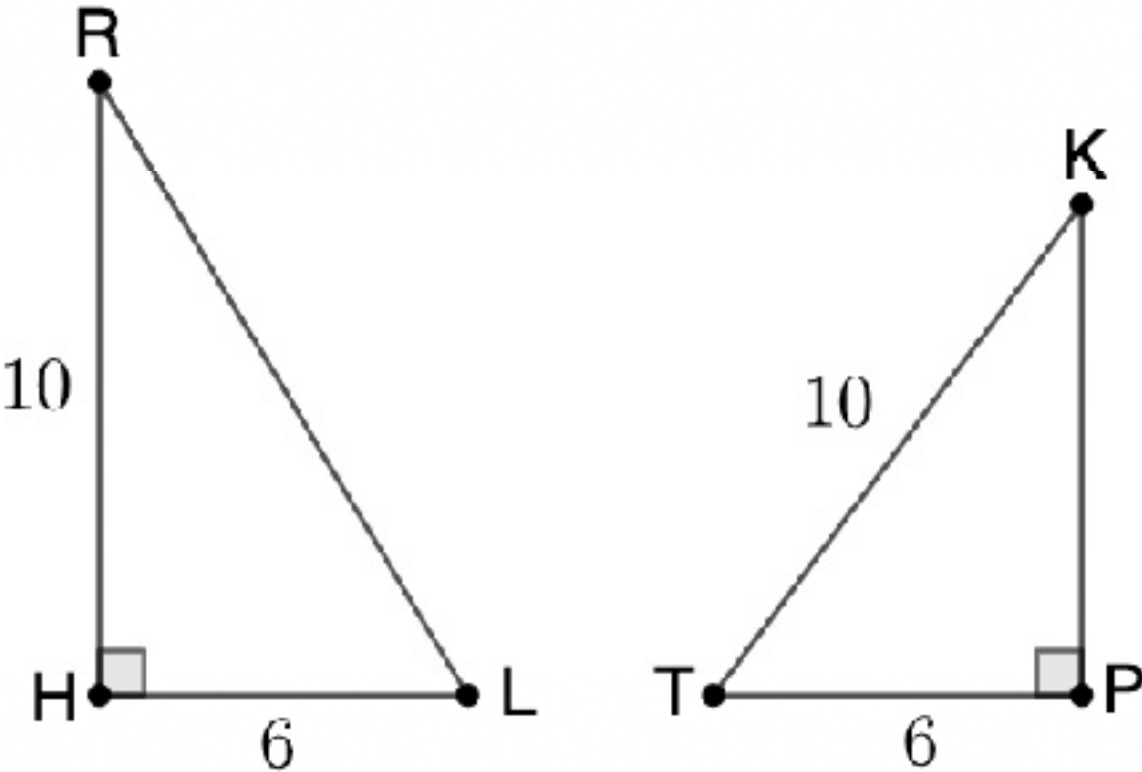
- ☐ hypotenuse
☐ leg

- ☐ a
☐ b
☐ c

$c = \sqrt{\quad}$
 $c \approx \underline{\hspace{2cm}}$

5.6.5 Collaboration: What’s the Leg?

1. Triangles RHL and TKP are shown. Both triangles have one side with an unknown measurement.



a. How are these triangles similar?

b. How are these triangles different?

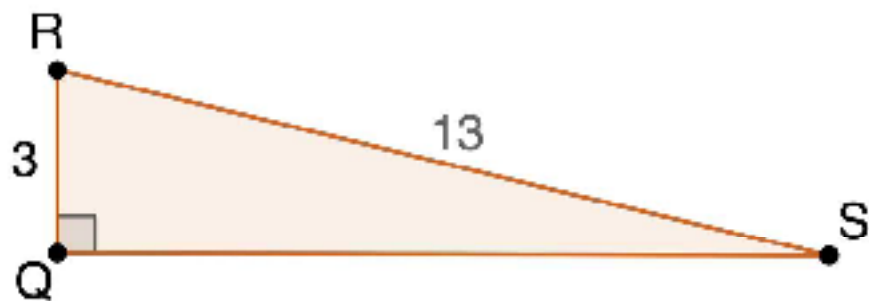
c. Work with your partner to write equations to determine the length of the missing side in each right triangle using the Pythagorean Theorem.

Triangle RHL	Triangle TKP

5.6.6 Guided Instruction: Solving for a Missing Leg Using the Pythagorean Theorem

The Pythagorean Theorem can also be used to find a missing leg in a right triangle.

1. Use the Pythagorean Theorem to find the approximate measure of $\overline{QS}$.



- a. State the Pythagorean Theorem.

- b. Substitute the length of the

○ hypotenuse

○ legs

 for c .

- c. Substitute the given leg length for

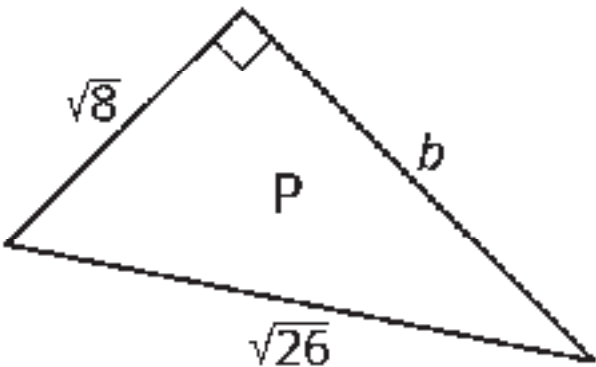
○ a or b .

○ a or c .

○ b or c .

- d. Solve the equation for the missing variable.

2. Find the exact measure of b in the diagram.



- a. State the Pythagorean Theorem.

- b. Substitute the length of the

○ hypotenuse

○ leg

 for c .

- c. Substitute the given leg length for

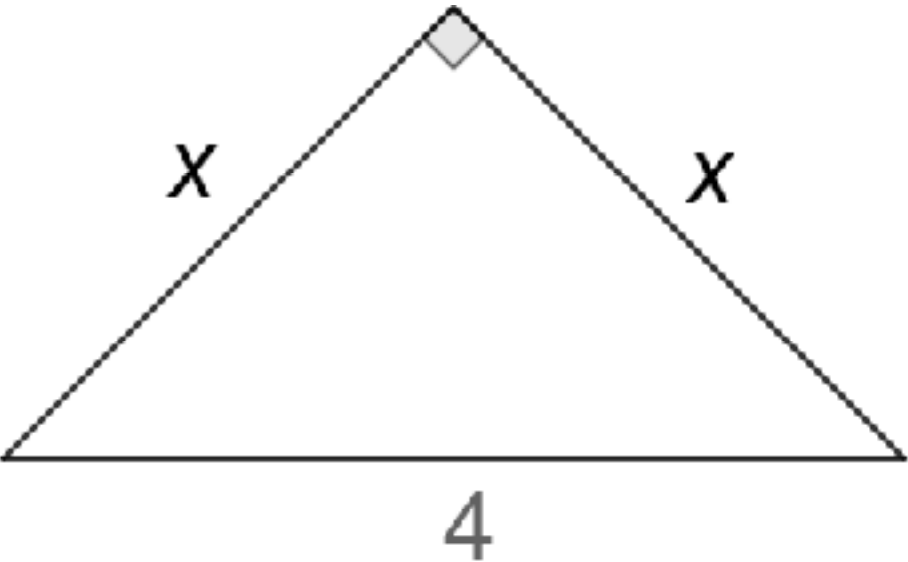
○ a or b .

○ a or c .

○ b or c .

- d. Solve the equation for the missing variable.

3. Find the exact length of x in the diagram using the Pythagorean Theorem.



a. State the Pythagorean Theorem.

b. Substitute the length of the

hypotenuse

leg

 for c .

c. Substitute the given leg length for

a and b .

a and c .

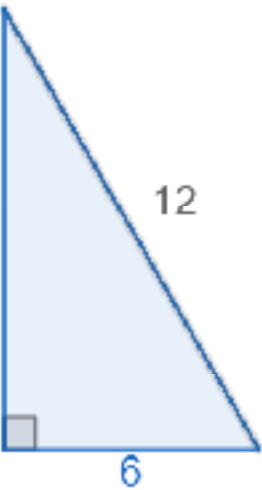
b and c .

d. Solve the equation for the missing variable.

5.6.7 Your Turn!

1. Match the triangle to the appropriate equation that could be used to solve for the missing side.

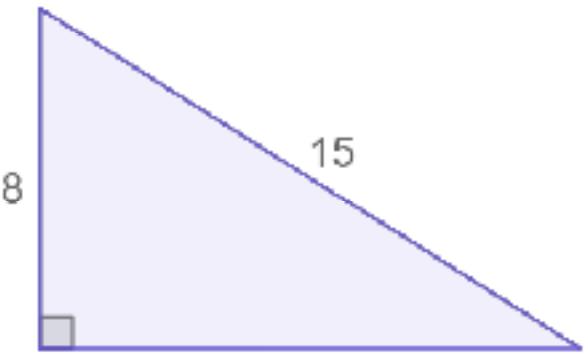
A. ____



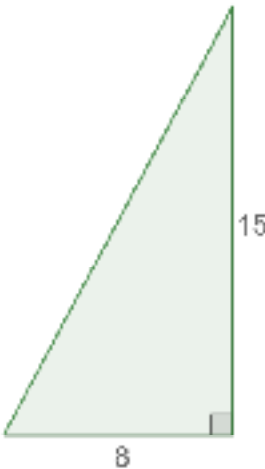
B. ____



C. ____



D. ____



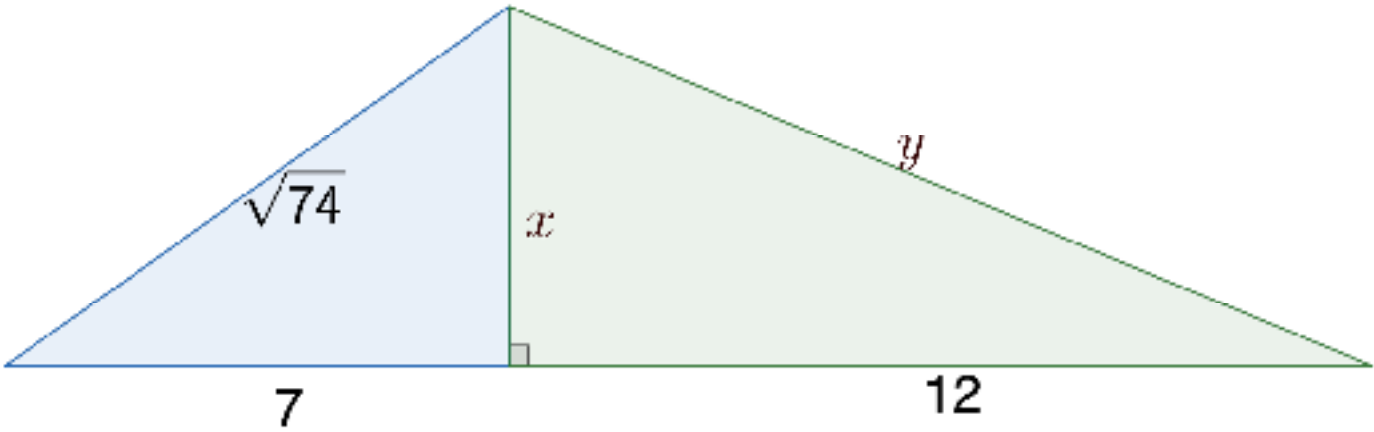
I. $8^2 + b^2 = 15^2$

II. $12^2 + 6^2 = c^2$

III. $a^2 + 6^2 = 12^2$

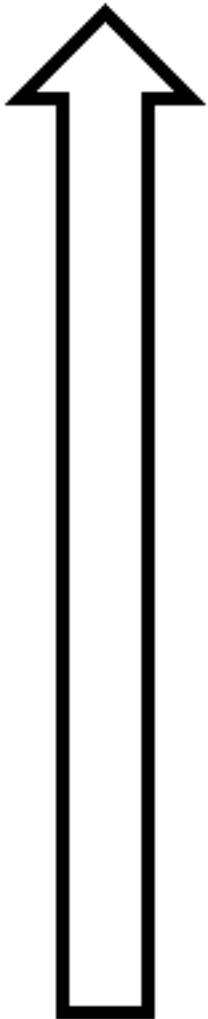
IV. $8^2 + 15^2 = c^2$

2. Find the measures of x and y in the diagram using the Pythagorean Theorem.



5.6.8 Wrap-Up: Reflection

1. Color in the arrow up to the statement which best describes your current understanding.



I'm so confident, I can explain this to someone else!

I can get to the right answer, but I don't understand well enough to explain it yet.

I understand some of this, but I don't understand all of it yet.

I tried hard and listened, but I am finding this challenging. I will make sure I get help with this next lesson.

I do not understand any of this yet. There are things I could do to be a better learner next lesson.

